

**Student:** Jonatan Rabjerg  
**Date:** 1/15/24

**Instructor:** Andreas Basse-O'Connor  
**Course:** Calculus Beta

**Assignment:** MyLab Math - Uge 10

Solve the initial value problem.

$$\frac{dy}{dx} + 3xy = 10x, \quad y(0) = 5$$

To solve the linear equation  $y' + P(x)y = Q(x)$ , multiply both sides by the integrating factor  $v(x) = e^{\int P(x) dx}$  and integrate both sides.

First identify  $P(x)$ .

$$P(x) = 3x$$

Now determine the integrating factor.

$$\begin{aligned} v(x) &= e^{\int P(x) dx} \\ &= e^{\int 3x dx} && \text{Substitute } 3x \text{ for } P(x). \\ &= e^{\frac{3x^2}{2}} && \text{Integrate.} \end{aligned}$$

Next, multiply both sides of the linear equation by the integrating factor.

$$\begin{aligned} \frac{dy}{dx} + 3xy &= 10x \\ e^{\frac{3x^2}{2}} \cdot \left( \frac{dy}{dx} + 3xy \right) &= e^{\frac{3x^2}{2}} \cdot 10x && \text{Multiply both sides by } v(x). \\ \frac{d}{dx} \left( e^{\frac{3x^2}{2}} \cdot y \right) &= e^{\frac{3x^2}{2}} \cdot 10x && \text{The left side is } \frac{d}{dx}(v \cdot y). \end{aligned}$$

Integrate both sides and simplify.

$$\begin{aligned} \frac{d}{dx} \left( e^{\frac{3x^2}{2}} \cdot y \right) &= e^{\frac{3x^2}{2}} \cdot 10x \\ e^{\frac{3x^2}{2}} \cdot y &= \frac{10}{3} e^{\frac{3x^2}{2}} + C \end{aligned}$$

Now solve for  $y$ .

$$\begin{aligned} e^{\frac{3x^2}{2}} \cdot y &= \frac{10}{3} e^{\frac{3x^2}{2}} + C \\ y &= \frac{10}{3} + C e^{-\frac{3x^2}{2}} \end{aligned}$$

Next, use the initial condition to solve for  $C$ .

$$\begin{aligned} y &= \frac{10}{3} + C e^{-\frac{3x^2}{2}} \\ 5 &= \frac{10}{3} + C e^{-\frac{3(0)^2}{2}} && \text{Substitute the values for } x \text{ and } y. \\ \frac{5}{3} &= C && \text{Simplify and solve for } C. \end{aligned}$$

Finally, substitute  $\frac{5}{3}$  for  $C$  to complete the solution.

$$y = \frac{10}{3} + Ce^{-\frac{3x^2}{2}}$$

$$y = \frac{10}{3} + \frac{5}{3}e^{-\frac{3x^2}{2}}$$